

Anonymity in Multi-hop Neighborhoods

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Universiteit
Leiden



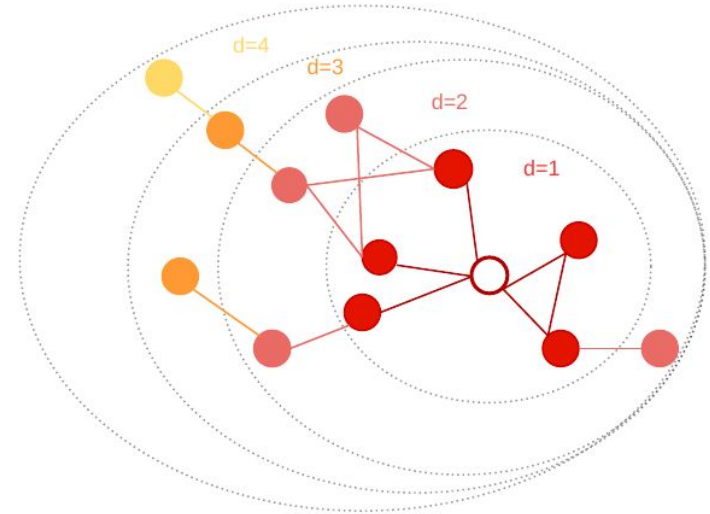
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Overview

- Anonymity in networks
- Approach 1
 - Multi-hop neighborhoods
- Approach 2
 - Cascading algorithm
- Conclusions

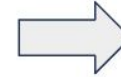
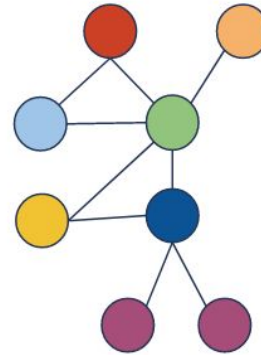


Anonymity in Networks

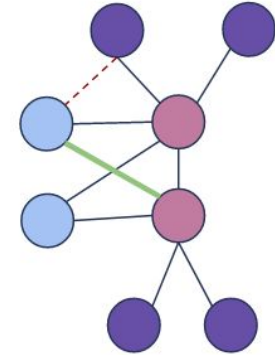
"Unlocking Network data"

- Sensitive information
 - **Relations**
 - Attributes
- Comply with privacy laws
 - **Identifiability of individuals**
 - Information about individuals
- For now:
 - **Anonymous = not unique**

unique node
≠ anonymous



non-unique nodes
= anonymous



Approach 1: Multi-hop Neighborhoods

Multi-hop neighborhood of a node v

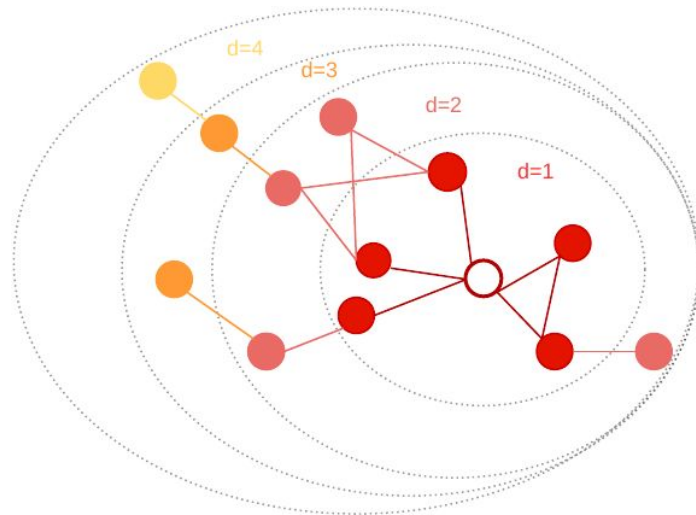
- All nodes up to distance d
- All connections between these nodes

Uniquely identifiable if:

- d -Neighborhood:
 - ☐ Is unique (isomorphism)
 - ☐ No node with same structural position (orbit)

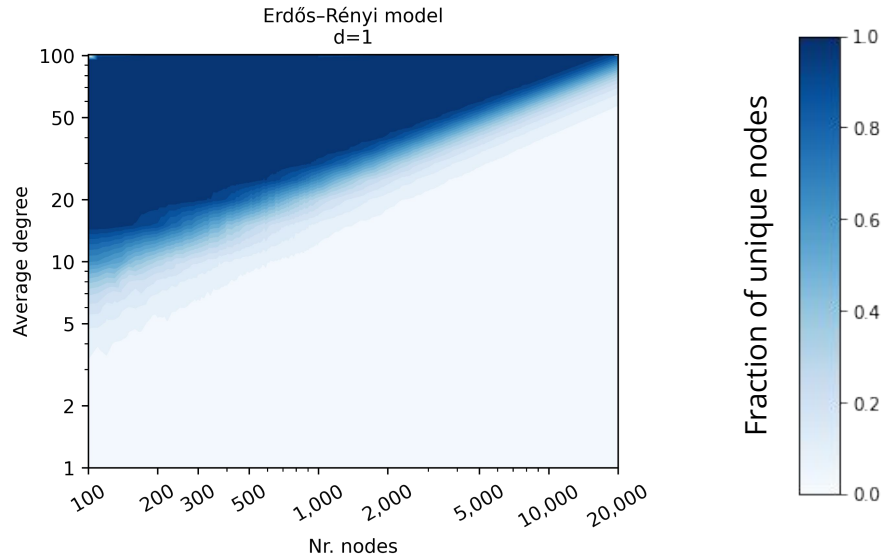
Which value of d to use?

- Information, computation time

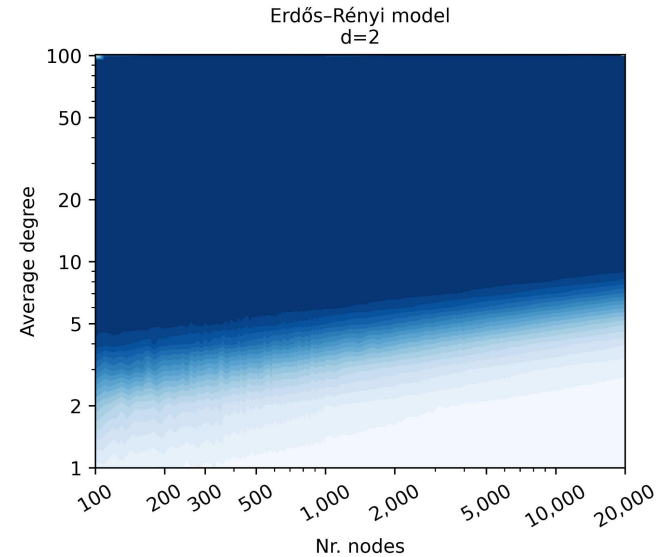
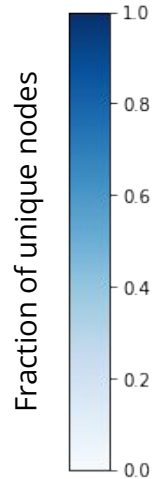
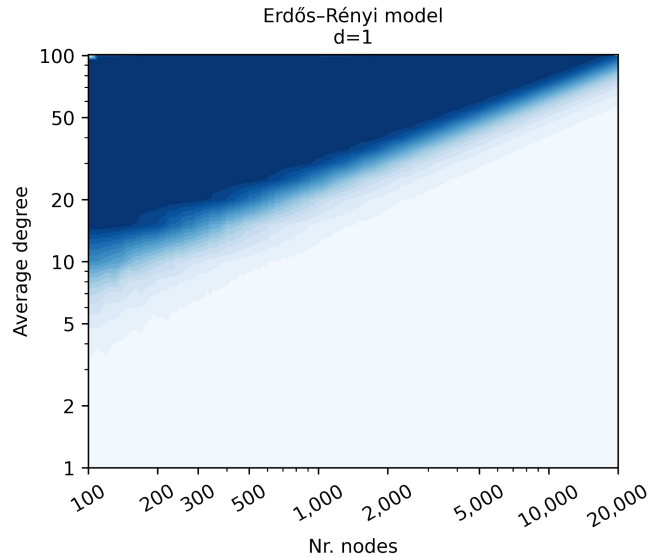


Code available: <https://github.com/Racheldelong/dkAnonymity>

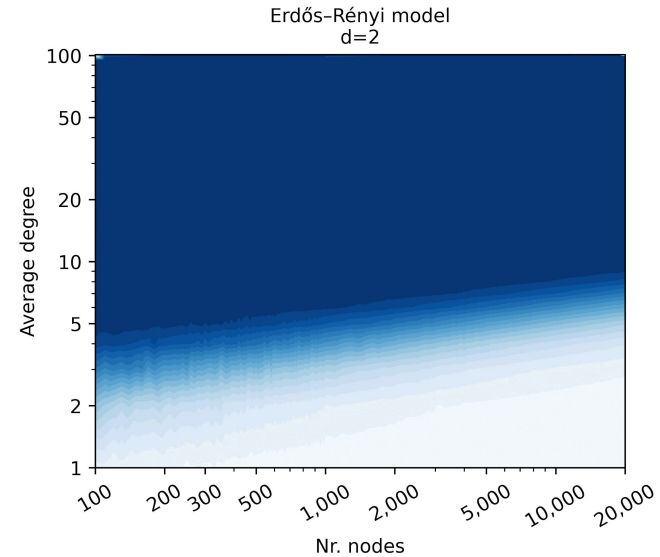
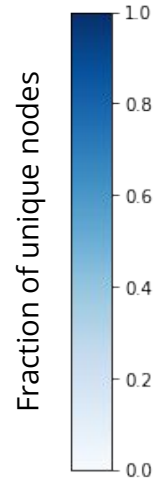
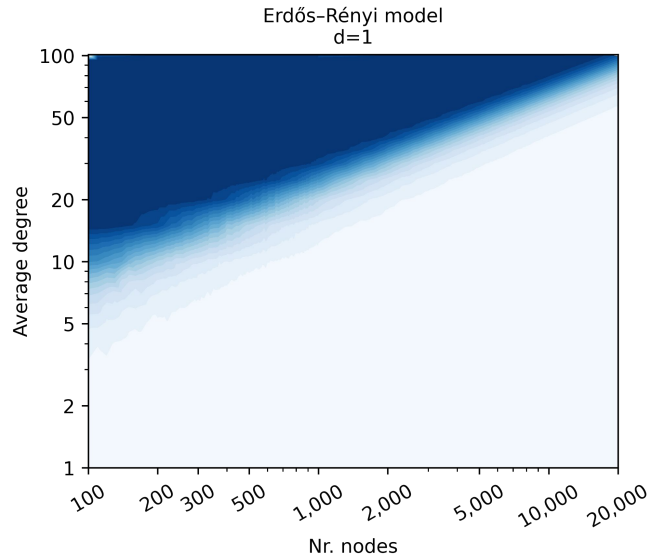
Approach 1: Results on Network Models



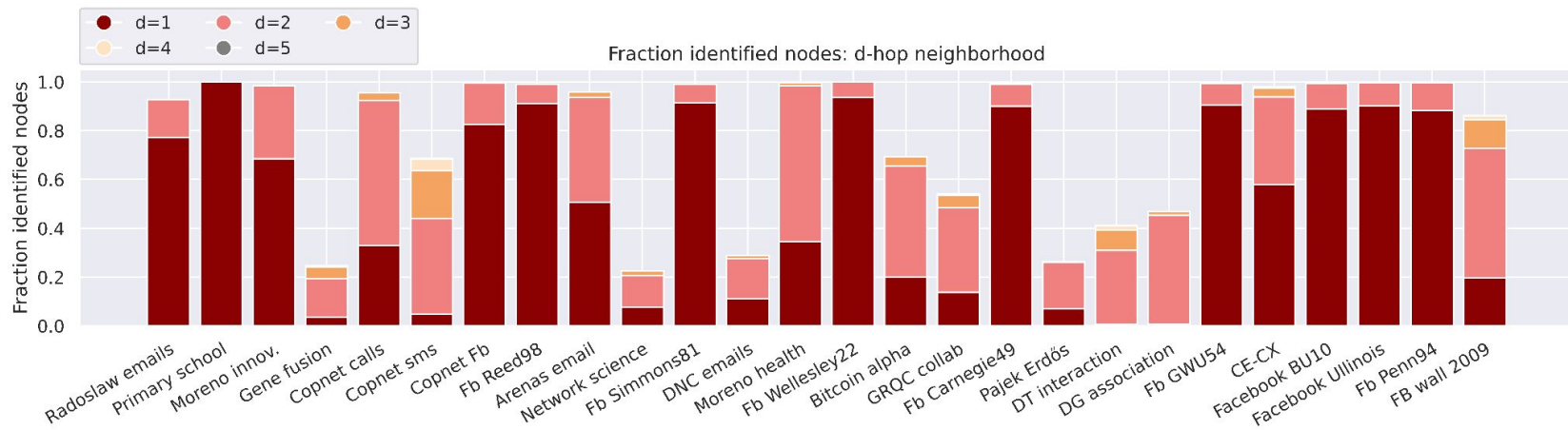
Approach 1: Results on Network Models



Approach 1: Results on Network Models



Approach 1: Results on Empirical Networks



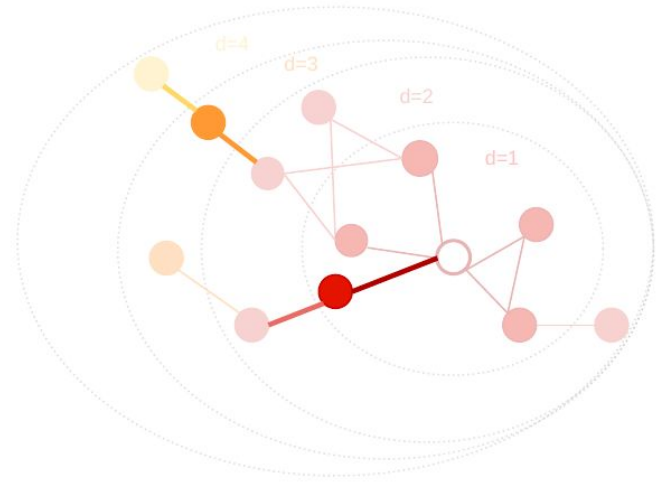
$|V|$: 167 - 46,951 $|E|$: 279 - 1,362,220

>100% increase: **12** / **26** networks when $d=1 \rightarrow d=2$

Approach 2: Cascading Algorithm

Knowledge:

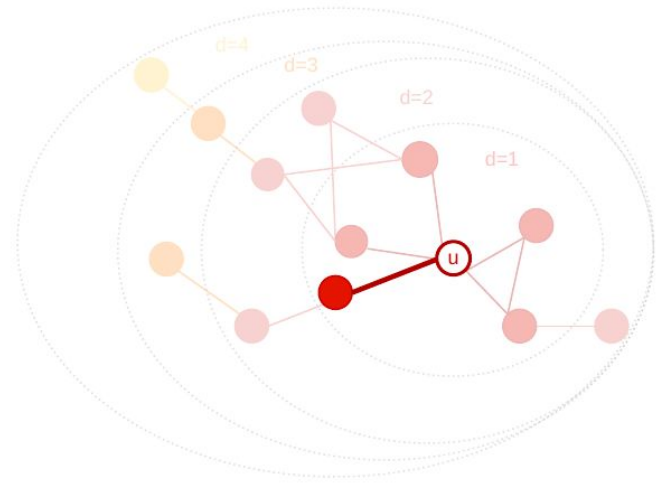
- 1-neighborhood



Approach 2: Cascading Algorithm

Knowledge:

- 1-neighborhood
- **Existence of a connection**



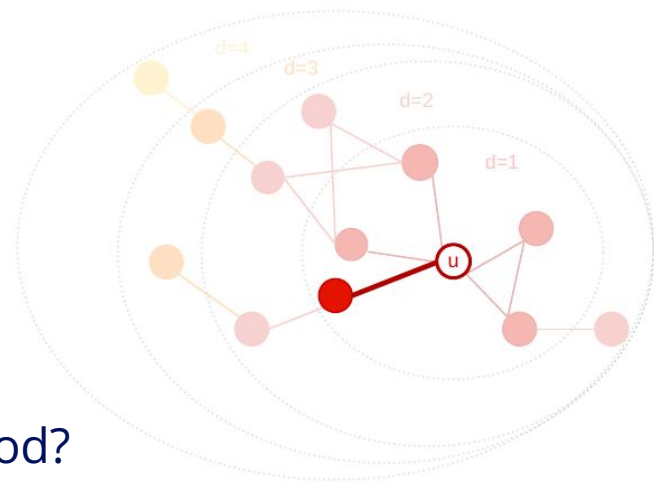
Approach 2: Cascading Algorithm

Knowledge:

- 1-neighborhood
- **Existence of a connection**

Algorithm:

- Find a uniquely identified node u ($d=1$)
- Is a neighbor v of u unique in this neighborhood?
 - Yes $\rightarrow$ uniquely identified



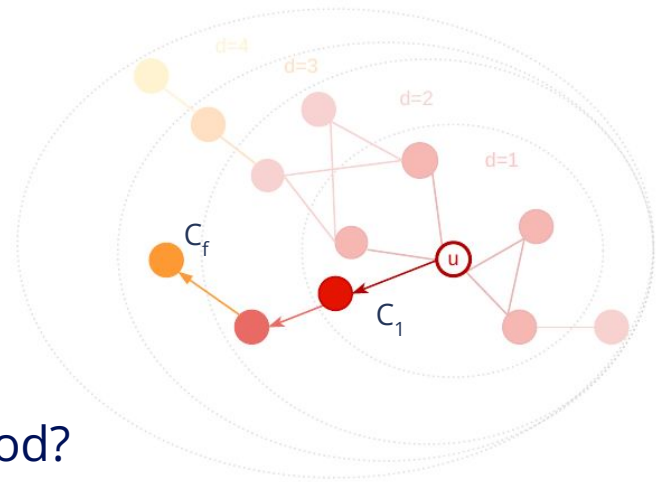
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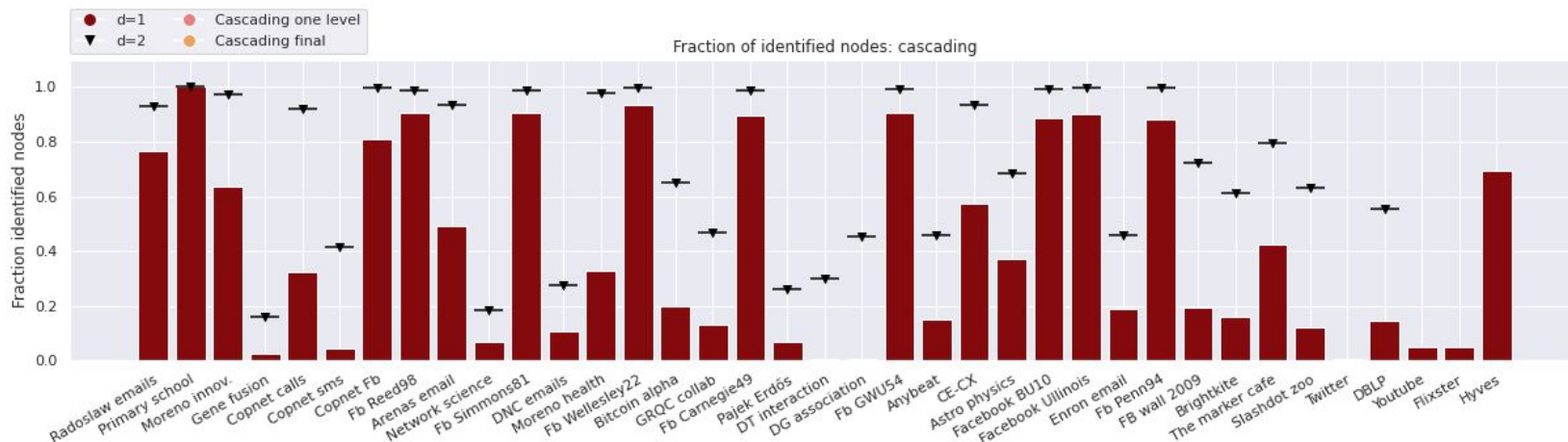
- 1-neighborhood
- **Existence of a connection**

Algorithm:

- Find a uniquely identified node u ($d=1$)
- Is a neighbor v of u unique in this neighborhood?
 - Yes $\rightarrow$ uniquely identified
 - Further levels: repeat for uniquely identified v

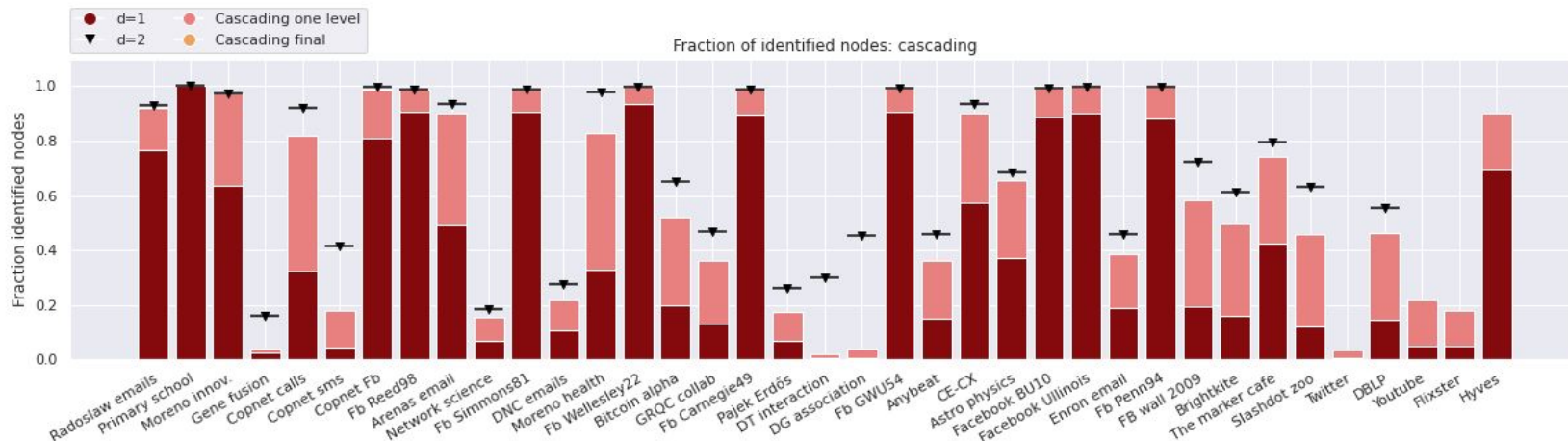


Approach 2: Results on Empirical Network Datasets



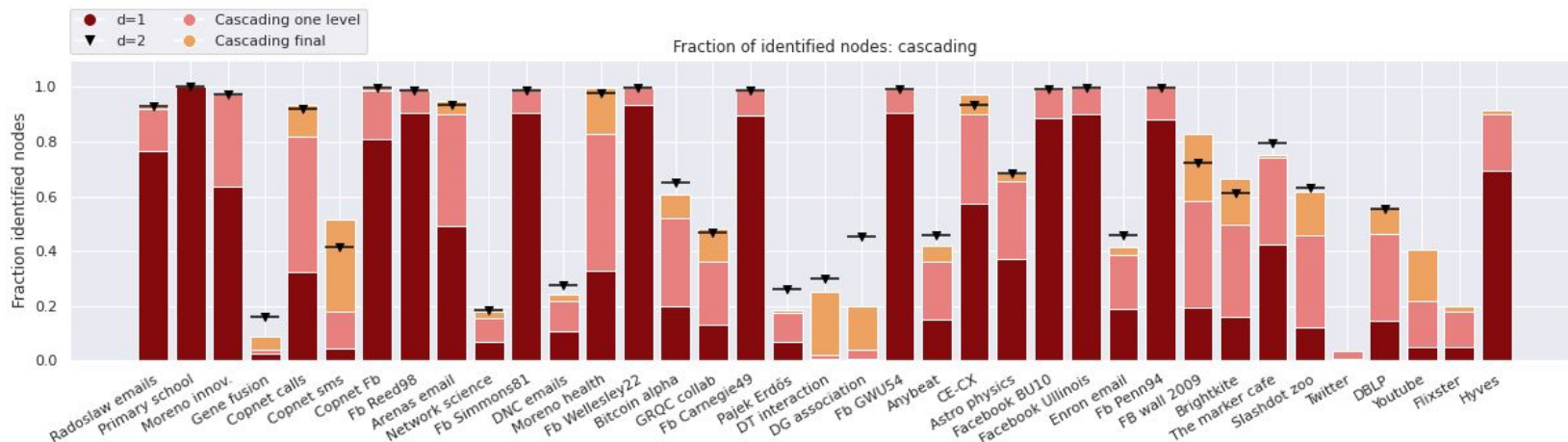
$|V|$: 167 - 8,113,018 $|E|$: 279 - 912,120,070

Approach 2: Results on Empirical Network Datasets



$|V|$: 167 - 8,113,018 $|E|$: 279 - 912,120,070
 >100% increase: 19 / 37

Approach 2: Results on Empirical Network Datasets



$|V|$: 167 - 8,113,018 $|E|$: 279 - 912,120,070

>100% increase: 19 / 3 / 37

Conclusion

- Knowledge about multi-hop neighborhoods
 - $d=1 \rightarrow d=2$ large de-anonymizing effect
 - >100% increase in uniquely identifiable nodes for many networks
- Knowledge about *one additional link* (cascading algorithm)
 - Often large de-anonymizing effect
 - Fast computation
- Future work
 - Anonymizing networks by means of perturbation

Contact

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Code: <https://github.com/Racheldelong/dkAnonymity>